

# Orca Bricks: Research & Ideas Sheets

FLL BIOGLOW 2026-27 Innovation Project · typed up October 4, 2026

Seven Orca Bricks team members filled in a Research & Ideas Sheet for the FLL BIOGLOW 2026-27 Innovation Project, most dated September 27, 2026. Every sheet names the same problem: deforestation, water pollution and air pollution. This is a typed copy of the handwritten sheets, with spelling cleaned up and uncertain words marked (?).

## Ideas at a glance

| # | Team member         | Big idea                                                                                       | Source                 |
|---|---------------------|------------------------------------------------------------------------------------------------|------------------------|
| 1 | Lydia               | A huge machine that plants native seeds; a traffic-light water pollution monitor with a filter | YouTube, Britannica    |
| 2 | Chansy J.           | Spread the word about pollution, person to person                                              | Balamurugan Thangamani |
| 3 | Krishiv             | Make little homes for animals                                                                  | PebbleGo               |
| 4 | Iraj                | A water monitor that triggers a purifier when toxic levels are too high                        | theroyalsociety.org    |
| 5 | Ishaani             | A robot trash can that comes to you when you need it                                           | royalsociety.org       |
| 6 | Shaurya             | Switch to natural gases as a different type of fuel                                            | None listed            |
| 7 | Devansh Molagavalli | A simple model showing how to help solve all three problems                                    | ChatGPT                |

Lydia and Iraj both proposed a water monitor paired with a filter or purifier.

# 1. Lydia

*Dated Sep. 27, 2026. Two pages, with a different idea on each side.*

**The problem we care about:** Deforestation, water and air pollution.

**Who does it affect? Why does it matter?** It affects animals and plants, and that can impact the ecosystem by taking away natural habitats.

**What we found out:**

- 1 There was a project to plant 1 trillion trees.
- 2 In South Korea they are doing reforestation.
- 3 If you plant the same trees to form an ecosystem, it won't work.

**Source / expert we asked:** YouTube, Britannica

## Idea 1: Native seed planter (front)

**Our big idea:** A huge seed planter machine that deposits native seeds.

**Sketch:** A wheeled machine with a wide row of planting teeth across the front.

**How it works:** It plants different native seeds into the ground to form an ecosystem.

**What we'll build to test it:** A lot of soil without any seeds.

**How we'll share it:** Social media.

**Questions we still have:** Left blank.

## Idea 2: Water pollution monitor (back)

**Our big idea:** A monitor that can tell people if the water is too polluted.

**Sketch:** A traffic light.

**How it works:** It is like a traffic light. It turns red when the water is too polluted, yellow when the water is a little polluted, and green when the water is clear. If the water is polluted, the water would go through a filter to filter out the germs.

**What we'll build to test it:** We will use a big bucket of water and pollute it.

**How we'll share it:** Social media.

## 2. Chansy J.

*Dated "September". Spelled Chansy as it reads on the sheet; spelling to be confirmed.*

**The problem we care about:** Air pollution, water pollution, deforestation all kill animals. How does it impact biodiversity?

**Who does it affect? Why does it matter?** It affects wildlife and us. It matters because if we don't act upon it the population will die.

**What we found out:**

- 1 Majority of ecosystem is underground.
- 2 Pollution is getting into animals.
- 3 Air pollution is bad for our lungs.

**Source / expert we asked:** Balamurugan Thangamani

**Our big idea:** We will tell everyone how air pollution etc. affects us and why it's important, and tell them to tell everyone they know.

**Sketch:** Two stick figures, one talking to the other.

**How it works:** After we tell our friends, they'll tell people to tell their friends, and so on.

**What we'll build to test it:** "Nothing 2 build"

**How we'll share it:** Word of mouth, texting.

**Questions we still have:** What if no one cares? What if word stops and everyone forgets?

### 3. Krishiv

*No date on the sheet.*

**The problem we care about:** Deforestation, water pollution, air pollution.

**Who does it affect? Why does it matter?** Animals, us.

**What we found out:**

- 1 3,000 animals die.
- 2 More animals are expected to die.
- 3 It affects us too.

**Source / expert we asked:** PebbleGo (also written beside points 1 and 2).

**Our big idea:** Make little homes.

**Sketch:** A stick figure next to a small house.

**How it works:** It is a small home.

**What we'll build to test it:** Home.

**How we'll share it:** Go to a zoo.

**Questions we still have:** How much money?

## 4. Iraj

*Dated 9/27.*

**The problem we care about:** The problem I care about is deforestation, water pollution, and air pollution.

**Who does it affect? Why does it matter?** It affects wildlife such as animals, plants, and trees. It matters because if we did not stop, we would end up getting ourselves extinct, because there will be no food or oxygen left for us.

**What we found out:**

- 1 Almost 30% of animals on Earth are endangered or basically extinct.
- 2 Some places on Earth are preserved for animals.
- 3 "CSAs (?) are helpers": growing multiple crops helps soil thrive. The first word is hard to read.

**Source / expert we asked:** [theroyalsociety.org](http://theroyalsociety.org)

**Our big idea:** The monitor will go off if the toxic level is too high, and the purifier will clean the water.

**Sketch:** Dirty water flowing into a water purifier with clean water below, and a monitor at the side.

**How it works:** When the monitor goes off, the water will flow through the purifier and get cleaned for drinking.

**What we'll build to test it:** A Lego pool with a retractable hole.

**How we'll share it:** Raise awareness that we are getting closer to being extinct if we pollute water and air and destroy forests.

**Questions we still have:** How to clean air.

## 5. Ishaani

*Dated Sep. 27. Two pages: the sheet and a back page with more research detail.*

**The problem we care about:** Deforestation, water pollution, and air pollution.

**Who does it affect? Why does it matter?** If we never had forests, water or air, none of the humans, animals, and plants would survive.

**What we found out:**

- 1 There are species that we don't know yet and are already going extinct.
- 2 After people started making new inventions, biodiversity was going less and less.
- 3 People have tried to help biodiversity, but every year it gets worse and worse.

**Source / expert we asked:** royalsociety.org

**More on the research (back page):**

- Point 2: After people started making new plant-based inventions like gases and new fertilizers, it can harm habitats and animals.
- Point 3: People have tried to help biodiversity by cleaning up lakes, beaches and rivers. They've also tried to stop using pesticides, but the lakes, beaches, and rivers get dirty again and pesticides are still being used over compost.

**Our big idea:** Build a robot that knows when someone needs a trash can and comes so the person can throw the trash away. We could also walk or ride bikes or scooters instead of cars.

**Sketch:** People outdoors near trees, with a robot trash can on wheels rolling up to them.

**How it works:** When someone needs to throw away something, the robot comes so the person can throw it away.

**What we'll build to test it:** Grab different pieces of trash and put them around the beach to see if it'll pick them up.

**How we'll share it:** Unfinished (only "Sh" is written).

**Questions we still have:** If we create AI robots, won't they destroy nature while trying to help us?

## 6. Shaurya

*No name on the sheet. Dated 6/27/26, probably meant as 9/27/26.*

**The problem we care about:** Deforestation, air pollution, and water pollution.

**Who does it affect? Why does it matter?** It affects the animals that will eventually go extinct.

**What we found out:** Left blank, with no source listed.

**Our big idea:** My idea is to switch to natural gases.

**Sketch:** A truck at a gas pump, crossed out, next to a truck at a different kind of station.

**How it works:** Use a different type of fuel.

**What we'll build to test it:** Test different fuels.

**How we'll share it:** Talk to politicians.

**Questions we still have:** Will it work?

## 7. Devansh Molagavalli

*No date on the sheet.*

**The problem we care about:** The problems we care about are air pollution, deforestation and water pollution.

**Who does it affect? Why does it matter?** It means no trees and bad air pollution, and the sea animals will die.

**What we found out:**

- 1 Air pollution can harm plants, animals and insects, damage forests, and contribute to climate change.
- 2 Animals lose their home, plants disappear, and some species may become endangered or lose habitats.
- 3 Fish, turtles, birds and other aquatic organisms can be injured, poisoned... (the writing runs off the edge of the page).

**Source / expert we asked:** ChatGPT

**Our big idea:** Our idea is to create a simple model that shows how we can help solve three major environmental problems: air pollution, deforestation and water pollution.

**Sketch:** A sun-style diagram with a circle in the middle and labeled rays: save animals, grow trees, save energy, don't cut trees, save water, recycle, stop pollution, don't litter.

**How it works:** Faint and erased, possibly "build a bird house" (?).

**What we'll build to test it, how we'll share it, questions we still have:** Left blank.